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VeriSpoof: Beyond Real or Fake with One Audio LLM

One open audio language model for speech deepfake detection, source verification and grounded reasoning

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Models Benchmark Code Demo Project page

Abstract

A speech deepfake detector reports how likely a recording is synthetic, but a forensic analyst also asks which system produced it, whether two recordings share a source, and on what evidence the conclusion rests. We present *VeriSpoof*, an audio language model that unifies five forensic tasks — deepfake detection, generator attribution (generator, acoustic model and vocoder), recording-condition analysis, synthesized-span localisation and source verification — with one set of weights, and supports each decision with grounded rationales whose every statement can be independently verified. Every answer is a sentence whose likelihood is read out as a score, so detection and source verification need no separate heads and can be thresholded at an operating point. Every explanation consists of statements with an independent check: observations are measured on the waveform the model hears, synthesized spans come from labels, and the source profile from generator metadata. Counterfactual examples teach the decision to follow the stated evidence. On the 14 test sets of Speech-DF-Arena, *VeriSpoof-Plus* reaches 2.14% pooled EER, third of 28 systems and first among open-weight ones, and *VeriSpoof-Base*, trained on the official training splits of eight corpora only, reaches 4.43%, less than half the EER of the next open-weight detector. With the same weights, *VeriSpoof-Plus* decides whether two recordings from generators unseen in training share a source at 6.2% EER. Controlled studies show that the audio representation decides what is detected, the LLM how well two recordings are compared, and that source supervision is necessary for neural-codec fakes. Rationales can be checked statement by statement, catch partial fakes that the one-step answer misses (99.6 against 78.4% on PartialEdit), and determine the decision under edits never seen in training (99.5%). Only 29.7 M of the 2.85 B parameters are trained, so training fits a single 24 GB GPU with gradient checkpointing. Both models, the inference code, a demo, and *VeriSpoof-Bench*, the evaluation suite of all these questions, are released.

2.14%
pooled EER on 14 Speech-DF-Arena sets: 3rd of 28, 1st with open weights

6.2%
EER on same-source pairs from generators unseen in training

99.6%
PartialEdit fakes caught after reasoning; the one-step answer catches 78.4%

99.5%
decisions that follow an edit of the stated evidence never seen in training

1 Introduction

Speech deepfakes pose a significant security threat nowadays. A suspicious speech recording can raise several related questions: whether it is synthetic, which system produced it, and whether another recording came from the same source. Consider a voice message submitted as evidence. A deepfake detector can tell an analyst that it is probably synthetic, but the analyst's next questions are harder: which system produced it, whether it came from the same system as another message in the case, and on what grounds the detector believes so. The answers link a recording to a tool, and through the tool to a person or a campaign; a conclusion that cannot be explained cannot be contested.

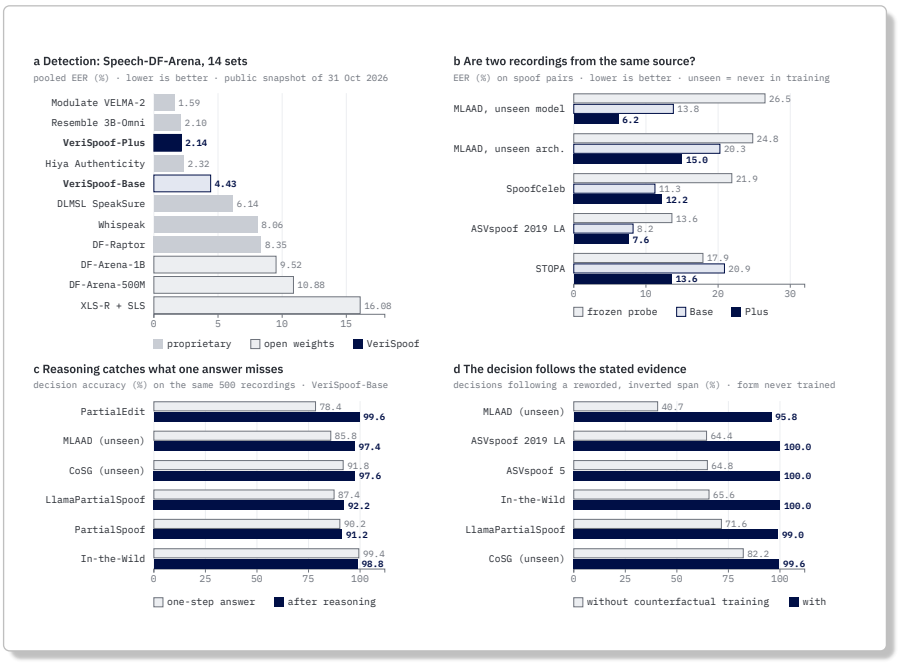


Figure 1. VeriSpoof at a glance. (a) Detection on the 14 test sets of Speech-DF-Arena [1], pooled EER (one threshold over all sets, the board's ranking); the public snapshot of 31 Oct 2026 with both VeriSpoof models inserted, top 11 shown. (b) The same weights decide whether two spoofed recordings come from the same generator, against a pair probe on the frozen anti-spoofing encoder DF-Arena-1B (§6.2). (c) Accuracy of the one-step answer and of the decision VeriSpoof-Base reaches after writing its rationale (§6.7). (d) Share of decisions that follow a localization replaced by its opposite, in a wording never used in training (§6.8).

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Research has made the first question tractable: detectors trained on large and diverse corpora reach low error rates across many benchmarks [1–3]. Source attribution, verification, and explanation have also been studied. For a forensic workflow, open detectors leave three problems.

1. **One score answers one question.** Source tracing assigns a recording to a closed set of known systems or attributes [4, 5], while comparing two recordings needs a separate Siamese network [6, 7]. Each further question costs another model, trained on its own data.
2. **Explanations cannot serve as evidence.** Audio language models (ALMs) for deepfakes add explanations [8–10], but these are free text written or distilled by a teacher model. Whether a stated observation is true of the recording is not checked, and whether the decision depends on it is not tested. Such an explanation can be fluent and wrong, or correct and irrelevant to the decision [11, 12].
3. **It is unclear where to invest.** Speech-recognition encoders have been argued to discard the artifacts of synthesis [13], and source labels are often assumed to help detection [8] without controlling for the number of authenticity labels or training steps. Which part of an ALM limits detection, and which limits comparing recordings, has not been measured.

VeriSpoof addresses all three problems in a single model: it unifies detection, attribution, recording-condition analysis, localisation and source verification, and accompanies every decision with rationales that can be checked statement by statement against the signal. For the first, it casts every forensic question as a question with candidate answers written as sentences and reads the likelihood of each answer as a score, so one model detects spoofs, names the generator, its acoustic model and vocoder, describes the recording channel and verifies whether two recordings share a source (Figure 3). For the second, it writes rationales in fixed fields, each with a target computed from the exact audio the model hears, so that every statement has a check without trusting the model. Training on counterfactual rationales further ensures that the decision follows the stated evidence (Figure 2). For the third, we run controlled studies that change one component or one form of supervision at a time on the same test trials, measuring which part of the model limits detection and which limits source comparison.

Key results

- **Best open-weight detector.** On the 14 Speech-DF-Arena sets, *VeriSpoof-Plus* reaches 2.14% pooled and 1.79% mean EER (third of 28 systems on the public snapshot), and *VeriSpoof-Base*, trained on 0.38 M utterances of official training splits, 4.43%; the best other open-weight system reaches 9.52% (§6.1).
- **The same weights verify sources.** For pairs from generators unseen in training, *VeriSpoof-Plus* decides whether they share a source at 6.2% EER for new models of known architectures and 15.0% for new architectures, against 26.5 and 24.8% for a pair probe on a frozen anti-spoofing encoder; general audio LLMs answer near chance (§6.2).
- **Representation decides detection, the LLM decides comparison.** With the encoder fixed, how its features are used lowers ASVspoof 5 EER from 9.6 to 3.1% while source verification stays at 16–19%; swapping the LLM moves detection by 0.5 points and source verification by 5.5 (§6.3).
- **Source labels are necessary for neural-codec fakes.** At a matched number of authenticity labels, source supervision helps on 8 of 16 sets and hurts on one; with five times more authenticity labels instead, CodecFake+ EER stays at 12.6 against 2.2% (§6.5).
- **Checkable, useful and faithful rationales.** Statements are checked one by one (SNR and level correct for 88–100% of recordings); the decision after reasoning catches partial and unfamiliar fakes the one-step answer misses (99.6 vs. 78.4% on PartialEdit); and with counterfactual training it follows a reworded, inverted localization in 99.5% of cases, against 73.8% without (§6.6–6.8).
- **Light to train and run.** Only 29.7 M of the 2.85 B parameters are trained. With gradient checkpointing, every training stage fits a single 24 GB GPU, and we estimate that training fits a 12 GB GPU and inference an 8 GB GPU (§6.9).

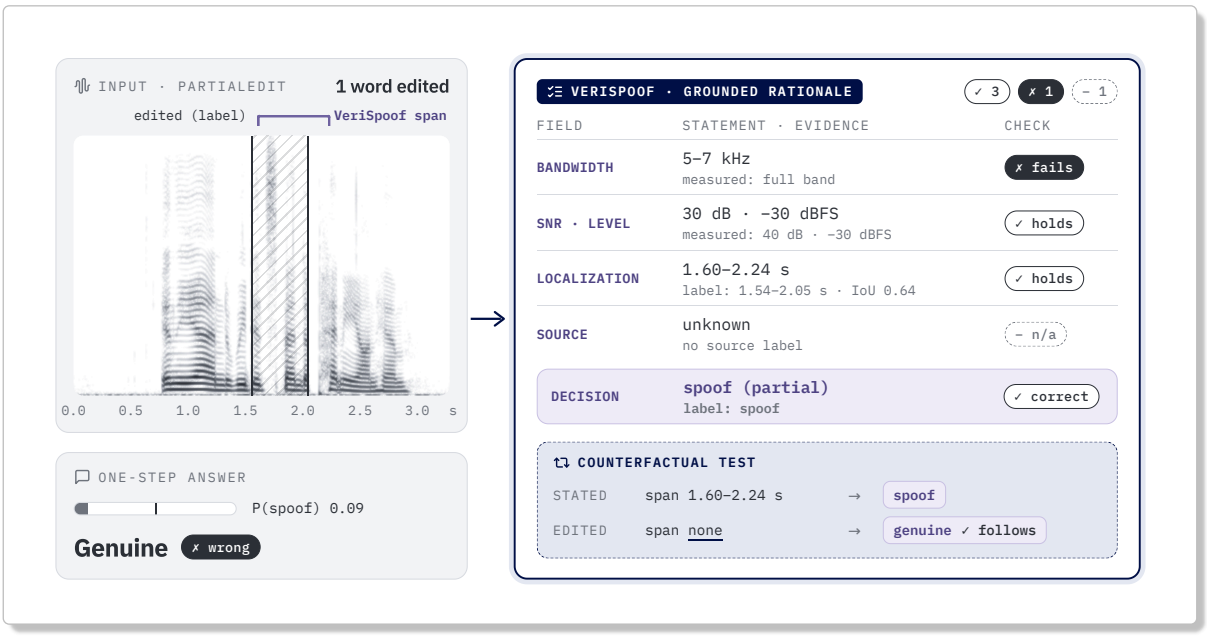


Figure 2. One score versus checkable evidence on the same recording, in which one word of genuine speech was replaced by synthesized speech. Asked directly, the model judges the recording genuine (left). Asked to reason, *VeriSpoof-Base* states observations, a synthesized span and a source profile before deciding (right). Each statement is checked against a waveform measurement or a label (✓ holds, × fails, – no label), and replacing the stated span by "none" flips the decision. Statements are abridged; the full output is in Figure 11.

Both models are released: *VeriSpoof-Base*, trained on the official training splits of eight corpora, and *VeriSpoof-Plus*, which extends Base with twelve additional corpora and a larger balanced training mixture (§4), together with the inference code with a vLLM [14] backend and a hosted demo. We also release *VeriSpoof-Bench*, the evaluation suite of this report: 54 test subsets with 177,852 items covering detection, attribution, source verification, recording conditions, grounded reasoning, and faithfulness, with scoring scripts and the per-item outputs of both models.

2 Related work

Speech deepfake detection. Most detectors pair a self-supervised front-end with a light back-end [15–17]. Recent work improves generalization through spectral residuals [18], high-order feature interactions [19], style–linguistics mismatch [20], dedicated pre-training [13], or data composition at scale [2, 3]. Evaluation has moved from ASVspoof [21, 22] and In-the-Wild [23] to multi-set boards such as Speech-DF-Arena [1] and SpoofRadar [24]. These systems return one score per recording; *VeriSpoof* returns its detection score as the likelihood of its own answer and is evaluated on both boards.

Source tracing and verification. Source tracing assigns a recording to a known system or to attributes such as its acoustic model and vocoder [4, 5], on corpora with many generators such as MLAAD [25] and SpoofCeleb [26]. Dedicated Siamese or two-branch networks score whether two recordings come from the same generator [6, 7]. We pose this comparison to the model that also detects and explains, on open-set protocols with unseen generators or architectures.

Audio language models for deepfakes. HoliAntiSpoof unifies detection, method recognition and localization [8], ThinkOmni generates reasoning with localization [9], and ICLAD prompts an ALM with comparison examples [10]. However, general-purpose ALMs [27–30] are not trained for these tasks. Their rationales, where present, are free text from a teacher model and are not checked against the signal: ICLAD reports hallucinated explanations, and ThinkOmni's training rationales are written by a model shown the labels. SpeakerLLM [31] unifies speaker profiling, recording conditions and verification reasoning for speakers.

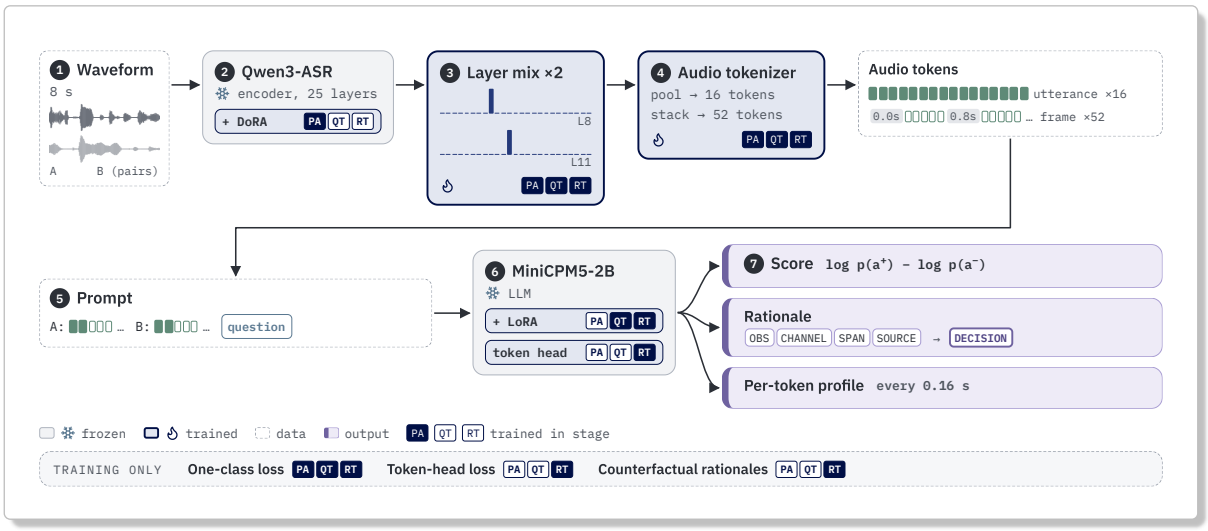


Figure 3. VeriSpoof. (1) An 8 s waveform, or two for a pair, enters (2) the Qwen3-ASR encoder, frozen apart from DoRA adapters. (3) Two learned mixes over its 25 layers feed (4) the audio tokenizer, which pools the recording into 16 utterance tokens and stacks pairs of frames with a text time marker every 0.8 s. (5) The prompt places the tokens of A, then those of B, before the question; (6) the frozen LLM with LoRA reads it and (7) returns a score, the log-likelihood difference of two answer sentences, a rationale whose decision comes last, or a per-token synthetic-speech profile. Grey: frozen; dark blue: trained by us, with the stages that train each part (PA: perception alignment, QT: forensic question tuning, RT: grounded rationale tuning).

Faithful explanations. In NLP, an explanation is faithful when the decision depends on it, which is tested by editing the rationale and observing the decision [11, 12, 32]. We apply such tests to audio, with edit forms held out from training, and train for them with counterfactual rationales.

3 VeriSpoof

3.1 Problem Formulation: Forensic Questions as Answer Scoring

Let u denote the input, either a single recording x or a pair (x, x'), and q a question with a set of candidate answers $\mathcal{A}_q$ each written as sentences. An ALM with parameters θ assigns each answer $a \in \mathcal{A}_q$, the likelihood $p_\theta(a \mid u, q)$ of its tokens. For a binary yes–no question, let a^+ and a^- denote the positive and negative answers; we read the score,

$$s_q(u) = \log p_\theta(a^+ \mid u, q) - \log p_\theta(a^- \mid u, q). \tag{1}$$

which can be evaluated by equal error rate (EER) and thresholded at a chosen false-alarm rate. Questions with more answers are scored by the accuracy of the most likely one. *VeriSpoof* answers four families of questions (Figure 5): *authenticity*; *attribution* of the generator, acoustic model or vocoder; *recording condition* (noise, reverberation, codec, post-processing); and *source verification* on pairs (same generator, acoustic model or vocoder?). No question has its own output head. For detection, for example, the question is "Is this recording genuine human speech or spoofed speech? Options: bona fide; spoof." and the two answers are full sentences ("This recording is a spoof: it is synthetic or converted speech, not genuine human speech."); the released models label a recording spoofed when $s_q(x) > 0$. Figure 10 shows the template of every question with an example answer.

Source verification is not the same as comparing attributes: two recordings from different systems can share every attribute class (e.g., generator family, acoustic model and vocoder), so a model that answers "same source" whenever the attributes agree fails on such pairs. Source-verification test sets therefore balance three strata: SAME, REVERSAL (different systems, every attribute class shared) and DIFF, so that attribute agreement alone does not reveal the answer.

3.2 Model

Perception in two resolutions. A Qwen3-ASR encoder [33] reads an 8 s window of the recording. A learned weighted sum of the hidden states of all 25 of its layers is passed to an audio tokenizer with two outputs: 16 utterance tokens, which carry what is constant across the recording and serve attribution and verification, and 52 frame tokens of 0.16 s, which preserve temporal information. Time markers every 0.8 s are inserted between frame tokens as text, so that the LLM can state spans in seconds. The encoder is adapted with DoRA [34] (r=16) during the first training stage only and frozen afterwards. For a pair, the tokens of both recordings are concatenated in the prompt.

LLM and read-out. The LLM is MiniCPM5-2B [35, 36], adapted with LoRA [37] (r=16, α=32) on the attention projections. The model has 2.85 B parameters, of which 29.7 M are trained (§6.9). The model is trained with next-token prediction on the answer tokens and answers are scored by Eq. (1). Two auxiliary losses support perception: a one-class authenticity loss [OC-softmax; 38] on the pooled audio tokens, which pulls bona fide speech towards a learned centre, and a per-token head that classifies each frame token as genuine or synthesized from the LLM's hidden state. The per-token head gives a synthetic-speech profile over time at no extra cost and serves as a reference for how much localization information the hidden states carry (§6.6). The model has two output modes: a one-step answer scored by Eq. (1), and a rationale followed by a decision, described in §3.3.

3.3 Rationales written from what the model hears

For forensic use, a decision must come with evidence that can be verified and contested, so *VeriSpoof* explains its decisions with rationales whose every statement can be checked independently of the model. A rationale $r = (r_1, \ldots, r_k; d)$ is a sequence of statements in fixed fields followed by a decision d. Each statement type τ has a checker $c_\tau(x, r_i) \in \{\text{true}, \text{false}\}$ that does not involve the model:

- OBSERVATIONS (bandwidth class, SNR, speech level, clipping, digital silence) are checked against measurements on the waveform;
- CHANNEL_STATUS (noise, reverberation, codec) against the logged recording condition;
- LOCALIZATION (synthesized spans in seconds, or none) against labelled spans;
- SOURCE_PROFILE (generation method, acoustic modelling, waveform generation) against generator metadata.

A rationale is *grounded* if every statement has a checker, so its correctness can be measured statement by statement. Channel statements describe the recording, not its authenticity, and are declared non-evidential. The data loader computes them directly from the exact window the model receives after augmentation: it measures observations on the waveform, takes channel statements from the augmentation log, maps labelled spans to window time, and reads the source profile from generator metadata. Each target is therefore true of its input, and the decision, written last, follows from the localization: a recording is labeled spoofed if and only if at least one synthesized span is stated, and genuine otherwise. The decision d is the final generated token sequence after the rationale fields; it is read as a score by Eq. (1) when thresholding is needed, or taken as the generated text directly. Verification rationales have the fields CHANNEL_STATUS, FINGERPRINT, PROFILE_COMPATIBILITY and DECISION; they state the fingerprint verdict before comparing attributes, because the attribute comparison can mislead the decision on REVERSAL pairs (§6.7). Figure 11 shows one output of each kind in the Appendix.

3.4 Three training stages

Training has three stages, each with its own trainable parts (Table 19).

1. **Perception alignment (PA)** trains the tokenizer and the encoder's DoRA adaptors on detection and source questions with the LLM frozen, so that the audio representation itself learns the relevant information before the LLM can compensate for weak audio tokens. The representation study of §6.3

is run at this stage for the same reason.

2. **Forensic question tuning (QT)** adds LoRA on the LLM and trains it on all perception questions, a form of multi-task instruction tuning: acoustic model, vocoder, source profile, recording condition, and pair questions on generator, acoustic model, vocoder and the authenticity of both recordings.
3. **Grounded rationale tuning (RT)** introduces detection and verification rationales and counterfactual examples at a lower learning rate. To preserve the one-step answering ability acquired in QT, the question-answering tasks of QT are replayed throughout this stage.

Each question has several paraphrases. Two models share this recipe and differ in their training data (§4): *VeriSpoof-Base*, the model of all controlled studies, and *VeriSpoof-Plus*. Counterfactual examples and their training procedure are described in §3.5.

3.5 Counterfactual training

Counterfactual training is applied in stage 3 (grounded rationale tuning). Grounded rationales alone do not make the decision depend on the rationale: the model can decide from the audio and write a matching rationale beside it. To make the decision follow the rationale, in 20% of the rationale examples we give the model a rationale whose localization is reversed (a labelled span replaced by "none", or "none" replaced by a span) and train only the decision that follows. The model is never taught to state something false about the audio. It is taught that the decision must follow the localization it is given. We further test faithfulness with interventions: a statement r_i is replaced by $\tilde{r}_i$ and the model decides again. The decision is faithful if it follows an edited localization ("no synthesized segment" $\Rightarrow$ genuine) and remains the same when an edited non-evidential statement is edited. Besides the edit form used in training, we test held-out forms: a paraphrased localization, an inserted span, and contradicted channel or observation statements.

3.6 Inference

One-step answers are scored by the likelihood of each candidate answer sentence; rationales are generated greedily. The released package exposes detection (`detect`), rationales with the per-token profile (`explain`), source verification at the generator, acoustic-model or vocoder level (`same_source`), verification rationales (`compare`), and closed questions on recording conditions and on generators from a list the user supplies. A vLLM [14] backend generates rationales from the same prompt embeddings (§6.9).

4 Training data and recipe

Figure 4 summarizes the training data of both models.

VeriSpoof-Base. Base uses the official training splits of eight detection corpora, ASVspoof 2019 LA [21], ASVspoof 5 [22], SpoofCeleb [26], MLAAD [25], STOPA [5], DFADD [39], LibriSeVoc [40] and CodecFake+ [41], and the partially spoofed corpora PartialSpoof [42], HAD [43] and PartialEdit [44] for rationales: about 0.38 M utterances. No evaluation split of any corpus is used. Base serves every controlled study of this report, so that its results are not tied to a particular choice of extra data.

VeriSpoof-Plus. Plus keeps the recipe and adds twelve corpora, spoofed speech from ASVspoof 2015 [45], ADD 2022 train / dev [46], Fake-or-Real [normalized version; 47], ReplayDF [48], SpeechFake [49], WaveFake [50] and Codecfake [codec conditions F01–F06; 51], and bona fide speech from a Common Voice 15.0 subset [52], LibriSpeech [53], AISHELL-3 [54], GigaSpeech [55] and CN-Celeb2 [56], 4.0 M utterances more. Its spoofed half is balanced over 211 sources (one generator of one corpus; MLAAD generators grouped into 82 architecture families) and its bona fide half over 18 datasets, so that the large added corpora do not dominate.

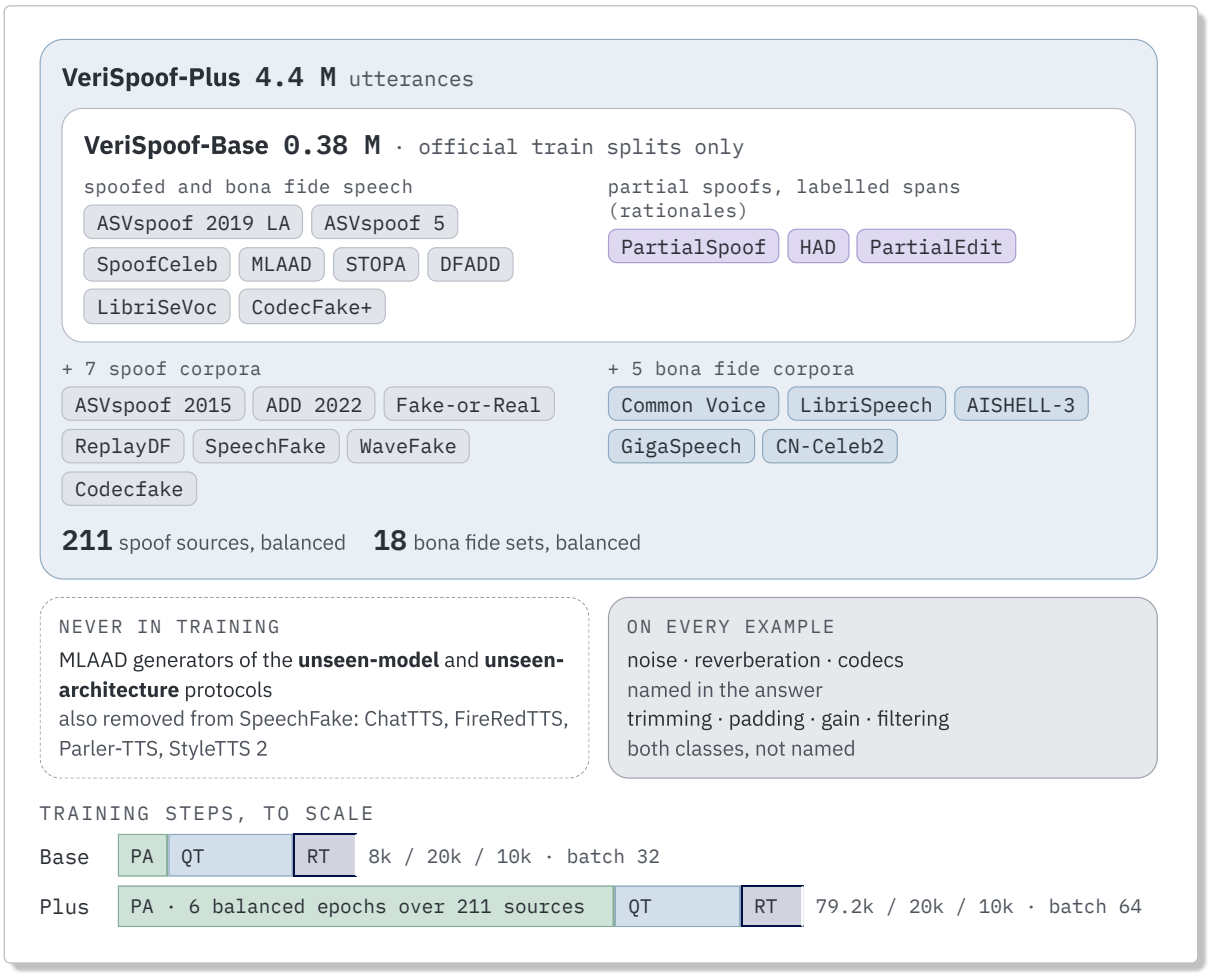


Figure 4. Training data. *VeriSpoof-Plus* keeps everything *VeriSpoof-Base* is trained on and adds twelve more corpora, balanced over 211 spoof sources and 18 bona fide datasets. Grey: corpora with spoofed speech; purple: partial spoofs whose labelled spans train the rationales; blue: bona fide speech only. Bottom: training steps of the three stages, perception alignment (PA), forensic question tuning (QT) and grounded rationale tuning (RT), to scale.

Held-out generators. The MLAAD generators of the unseen-model and unseen-architecture verification protocols are removed from all training data, including four SpeechFake systems that are among them (ChatTTS, FireRedTTS, Parler-TTS, StyleTTS 2). Codecfake and CodecFake+ are different corpora: CodecFake+ has a codec re-synthesis part ("CodecFake+" in our tables) and a part generated by codec-based speech generators (CoSG), whose generators are unseen in training.

Augmentation. Training copies carry labelled noise MUSAN [57], reverberation and codecs, including narrowband codecs, drawn anew for every example because the encoder runs inside the training loop. The recording-condition questions and the CHANNEL_STATUS statements are trained on these labels. Post-processing (silence trimming or padding, gain with clipping, high- or band-pass filtering) is applied identically to both classes, without naming it in the answer, which removes channel shortcuts (§6.4); on the corpora added for Plus, augmentation is applied without naming it either.

Recipe. Base trains 8k / 20k / 10k steps in the three stages at a global batch of 32 on one 48 GB GPU, about 14 GPU-hours (§6.9). Plus runs perception alignment for six balanced epochs (79.2k steps) at a global batch of 64 on two GPUs and keeps the step counts of the other two stages. Learning rates fall from 10^{-4} in perception alignment to 2×10^{-5} (LoRA) in rationale tuning. Questions are sampled by task weights; the full settings are in Table 19 and the training data in Figure 4.

5 Evaluation suite

Public detection boards. For detection we score every line of the 14 official protocols of Speech-DF-Arena [1]: ASVspoof 2019 LA (LA19), 2021 LA and DF (LA21, DF21) [58], ASVspoof 5 (ASV5), In-the-Wild (ITW) [23], Fake-or-Real (FoR), Codecfake (Codec), ADD 2022 tracks 1 and 3 (A22T1, A22T3), ADD 2023 rounds 1 and 2 (A23R1, A23R2) [59], DFADD, LibriSeVoc (LSV) and SONAR, 2.34 M trials in total. We report the pooled EER, which uses one threshold over all sets and decides the board ranking, and the mean over sets, and compare with the board's public snapshot of 31 Oct 2026. We also score the 16 sets of the SpoofRadar v1.0 protocol [24], which adds XMAD-Bench [60] in seven languages and SpoofCeleb, with its pooled metrics (Table 17). We did not submit to either leaderboard; our numbers are computed with the boards' own protocol files.

VeriSpoof-Bench. The other questions have no public benchmark, so we built one and release *VeriSpoof-Bench*. It has 54 subsets with 177,852 items in six groups (Figure 5, Figure 9): up to 10,000 items per closed-answer subset, drawn once and used for every model and checkpoint, so differences between columns of a table are paired. Source verification on MLAAD is split by how far the generators of a pair are from training: seen generators, unseen models of a seen architecture, and unseen architectures. Degraded copies come in two kinds: aug. copies apply noise, reverberation and codecs drawn as in training, and cond. copies carry one logged condition per recording, against which channel statements are checked. Rationales are evaluated on 500 utterances per set, and verification rationales on pairs balanced over the SAME, REVERSAL and DIFF strata. +proc marks copies with post-processing.

Metrics. Yes–no questions are scored by EER on the score of Eq. (1); questions with more answers by the accuracy of the most likely option. Observation statements are correct when the stated class matches the measured one (SNR within 10 dB, speech level within 5 dB, bandwidth and clipping classes). Localization is scored by the IoU of stated and labelled spans, and the per-token head by F1 of the synthesized class on the 0.16 s grid. Faithfulness is the share of decisions that follow an edited localization (follow rate) or stay unchanged under an edited non-evidential statement (stay rate).

Baselines. Detection is compared with the open-weight systems of Speech-DF-Arena and SpoofRadar; proprietary systems, whose training data are undisclosed, are listed for reference. No open system answers the source questions, so we train a pair probe on the frozen anti-spoofing encoder DF-Arena-1B with the Base training data, and prompt six general audio LLMs zero-shot with closed answer options [28–30, 61–63].

Controls. Controlled comparisons change one factor at a time with all else fixed and use the same test trials. The supervision study (§6.5) fixes its decision rule before the results and reports paired, class-stratified bootstrap intervals.

6 Results

6.1 Detection on public benchmarks

Best open-weight detector. On the 14 Speech-DF-Arena sets (Table 1), *VeriSpoof-Plus* has the lowest mean EER (1.79%) and pooled EER (2.14%) of all open-weight systems; the next best, DF-Arena-1B, reaches 5.66% and 9.52%, and every other open-weight system, including four listed only on SpoofRadar, has a mean EER above 5%. Inserted into the public snapshot, Plus ranks third of 28 systems by pooled EER, behind two proprietary systems, and Base fourth (Figure 1a). Base is trained on the official training splits of eight corpora; Fake-or-Real, Codecfake, ADD and SONAR have no training data in Base, and their EERs are its main gap to the systems above it. The score is the likelihood of the model's own answer, so answering in sentences costs neither a detection head nor accuracy.

SpoofRadar and unseen languages. On the 16 sets of the SpoofRadar v1.0 protocol (Table 2), both models have a lower pooled EER (Base 4.23%, Plus 4.04%) and higher accuracy and macro-F1 than



Figure 5. Questions and their test sets. Columns on the left are test splits of corpora used in training; columns on the right are sets never used in training. Filled circle: generators seen in training; open circle: generators, a vocoder pairing or a whole corpus unseen in training; dashed square: copies with added noise, reverberation and codecs (aug.), recorded under a logged condition (cond.) or post-processed (+proc). Speech-DF-Arena adds 14 public sets, scored on their full protocols. Figure 10 shows each question with an example answer.

Table 1. Detection on the 14 Speech-DF-Arena sets (EER, %; best per column in bold). Speech-DF-Arena snapshot of 31 Oct 2026; Teffic-Audio from its paper; ‡ from SpoofRadar, whose pooled EER covers other sets. Pooled: one threshold over all 14 sets. Systems: Modulate [64], Teffic-Audio [3], Hiya [65], DF-Arena [1], GRL [66], XLS-R + SLS [16], XLS-R + MultiConv [67], XLSR-Mamba [68], XLSR-MamBo [69], P3 [70], Nes2Net-X [17], AASIST [15].

System	ITW	LA19	LA21	DF21	ASV5	FoR	Codec	A22T1	A22T3	A23R1	A23R2	DFADD	LSV	SONAR	Mean	Pooled
Proprietary																
Modulate VELMA-2	1.27	0.30	1.33	0.33	**0.38**	0.13	1.54	**5.06**	1.17	1.04	**1.74**	**0.00**	0.27	0.89	**1.10**	1.59
Teffic-Audio	1.32	0.24	2.04	**0.26**	1.41	0.97	0.75	7.03	**0.84**	**0.36**	1.84	**0.00**	**0.00**	**0.25**	1.24	**1.45**
Hiya Authenticity v1	**0.67**	0.30	1.01	1.32	0.79	**0.00**	5.73	12.10	1.19	1.98	4.01	0.02	**0.00**	0.48	2.11	2.32
Open weights																
DF-Arena-1B	0.91	1.14	4.66	1.75	17.25	2.92	8.37	22.21	2.20	5.08	11.54	**0.00**	0.15	1.09	5.66	9.52
DF-Arena-500M	1.76	1.09	4.23	3.30	12.39	2.30	6.36	23.98	2.77	7.47	12.30	**0.00**	0.12	1.90	5.71	10.88
GRL‡	3.07	4.46	14.07	5.14	13.54	13.79	19.91	22.58	5.97	18.44	31.78	0.27	0.84	1.09	11.07	–
XLS-R + SLS	7.46	0.23	2.87	1.91	18.76	5.08	33.44	33.95	15.74	19.37	21.10	7.54	1.97	24.72	13.87	16.08
XLS-R + MultiConv‡	5.53	0.41	4.88	1.92	16.16	5.91	37.43	31.76	14.56	21.39	20.45	7.38	2.16	28.44	14.17	–
XLSR-Mamba	6.71	0.42	0.93	1.88	14.40	6.71	35.27	34.23	19.37	21.85	20.15	10.70	2.23	24.26	14.22	20.59
XLSR-MamBo‡	7.29	0.20	**0.88**	4.40	14.14	5.65	37.36	27.54	9.28	24.62	20.93	19.72	5.64	33.94	15.11	–
P3‡	5.86	1.73	15.48	5.87	20.03	19.92	26.89	39.73	14.30	25.98	29.03	4.90	0.65	7.58	15.57	–
Nes2Net-X	7.75	**0.12**	2.17	1.49	22.06	6.32	39.34	34.47	26.56	21.14	18.45	11.15	2.88	31.54	16.10	17.37
AASIST	43.01	0.83	11.46	21.07	35.53	21.64	51.06	47.92	33.19	47.76	32.47	41.86	38.02	57.47	34.52	32.77
Ours																
VeriSpoof-Base	1.20	0.49	5.48	0.97	2.59	5.39	4.97	14.05	3.87	5.01	10.61	**0.00**	0.02	5.19	4.28	4.43
VeriSpoof-Plus	1.90	0.22	3.44	1.24	3.40	0.09	**0.12**	8.65	1.18	0.50	2.41	**0.00**	0.02	1.92	1.79	2.14

every open-weight system on the published leaderboard, the best of which reaches 9.96%. XMAD-Bench [60], seven languages whose genuine speech comes from M-AILABS [71], MASC [72], VoxPopuli [73] and AISHELL-3, is unseen by both models apart from 1,001 AISHELL-3 recordings, and is detected by Base at 0.24% EER and by Plus at 1.33%.

For Plus, however, the pooled EER (4.04%) is far above the mean of its set EERs (1.65%), because one threshold does not fit every set. At the pooled threshold, 92.9% of the genuine Russian audiobook

recordings of XMAD-Bench (M-AILABS) are judged spoofed, as are 32.2% and 15.4% of the Spanish (M-AILABS) and Arabic (MASC) ones, although within each language the two classes are separated at $\leq$1.95% EER. Base misclassifies at most 3.1% of the genuine recordings of any language. The data added for Plus thus shift its scores on unseen genuine speech without making it less separable, a calibration cost that pooled metrics expose and per-set EER hides. A deployment on a new language or domain should therefore set its threshold on genuine speech of that domain.

Table 2. SpoofRadar v1.0 protocol (16 sets: the 14 of Table 1 with LibriSeVoc restricted to its test part, plus XMAD-Bench and SpoofCeleb). Pooled EER over all trials (%), and accuracy and macro-F1 (%) at the pooled-EER threshold, as defined by SpoofRadar; Avg.: mean of the 16 set EERs. Other rows: the published leaderboard (Leaderboard v1.0); ours computed with the same protocol files and labels from the source corpora. With one threshold and two classes, micro-F1 equals accuracy. Systems: Teffic-Audio [3]; Teffic Labs' closed-source systems built on WavLM-L [74], Whisper [75] and AASIST; DF Arena 500M [1], GRL [66], XLSR-SLS [16], XLSR-MamBo [69], XLSR + MultiConv [67], NaCL [76], TFCL [77], Nes2Net-X [17], XLSR-Mamba [68], P3 [70], W2V2-AASIST [78], RawNet2 [79], AASIST [15].

System	Pooled EER	ACC	Macro-F1	Avg. EER	XMAD	SpoofCeleb
Commercial						
Teffic-Audio	**1.37**	**98.63**	**98.12**	**1.09**	**0.02**	0.07
Closed source						
WavLM-L (Teffic)	3.65	96.35	95.06	2.88	0.19	0.48
Whisper (Teffic)	5.75	94.25	92.33	3.94	0.27	0.13
AASIST (Teffic)	11.52	88.48	85.20	7.55	2.28	1.55
Open weights						
DF Arena 500M V1	9.96	90.04	87.08	5.19	0.96	0.08
GRL	13.25	86.75	83.17	9.90	1.64	1.77
XLSR-SLS	15.97	84.28	80.25	14.07	9.13	24.52
XLSR-MamBo	16.33	83.67	79.64	15.17	13.58	17.53
XLSR + MultiConv	17.05	82.95	78.83	14.25	9.36	20.22
NaCL	17.22	82.78	78.64	16.83	14.11	22.81
TFCL	17.84	82.16	77.95	17.15	16.31	25.29
Nes2Net-X	17.97	82.03	77.81	18.23	10.51	54.97
XLSR-Mamba	19.47	80.53	76.16	18.91	15.84	27.58
P3	19.49	80.51	76.13	15.18	9.05	15.92
W2V2-AASIST	20.25	79.76	75.31	18.94	21.50	27.16
RawNet2	33.36	66.64	61.85	36.77	29.81	57.10
AASIST	33.66	66.34	61.56	36.08	48.37	58.26
Ours						
VeriSpoof-Base	4.23	95.77	94.29	3.76	0.24	0.20
VeriSpoof-Plus	4.04	95.96	94.54	1.65	1.33	**0.04**

What more data changes. From Base to Plus, 9 of the 14 common sets improve and 3 get worse. Gains are largest where Plus added training data of the same corpus: Codecfake 4.97 $\rightarrow$ 0.12 (Codecfake train), Fake-or-Real 5.39 $\rightarrow$ 0.09 (for-norm train), ADD 2023 rounds 1 and 2 5.01 / 10.61 $\rightarrow$ 0.50 / 2.41 (ADD 2022 train). Losses are on ASVspoof 5 (2.59 $\rightarrow$ 3.40), ASVspoof 2021 DF (0.97 $\rightarrow$ 1.24) and In-the-Wild (1.20 $\rightarrow$ 1.90): ASVspoof 5 shrinks from half of the training data to 8 of 211 balanced sources. The data mixture decides what the model is best at.

Degraded copies and the cost of rationales. On the *VeriSpoof-Bench* (Table 3), Plus is better on 12 of 16 rows and, as on the public sets, worse on ASVspoof 5 (2.23 $\rightarrow$ 3.41) and In-the-Wild (1.25 $\rightarrow$ 1.83); its largest gains are on degraded copies (LibriSeVoc aug. 10.31 $\rightarrow$ 7.72, LA19 cond. 5.00 $\rightarrow$ 3.87), where more varied training channels help. Between question tuning and rationale tuning, detection

changes by less than 0.3 points on most sets, so adding rationales does not cost one-step detection; the exceptions are ASVspoof 2021 LA (3.40 → 5.17) and CodecFake+ (2.24 → 3.95), neither of which supplies rationale data.

Table 3. Deepfake detection EER (%) on the *VeriSpoof-Bench*: Base after forensic question tuning (QT) and after grounded rationale tuning (final), and Plus. Up to 10,000 trials per set, the same trials for every column; full-set results on the public protocols are in Table 1.

Test set	Base (QT)	Base	Plus
ASVspoof 5	2.39	**2.23**	3.41
In-the-Wild	1.51	**1.25**	1.83
ASVspoof 2021 LA	3.40	5.17	**3.18**
ASVspoof 2021 DF	1.14	1.07	**0.76**
ASVspoof 2019 LA	0.47	0.56	**0.18**
SpoofCeleb	0.30	0.20	**0.10**
CodecFake+ CoSG	4.51	4.84	**2.34**
CodecFake+	2.24	3.95	**0.65**
DFADD	**0.00**	**0.00**	**0.00**
LibriSeVoc	**0.00**	**0.00**	**0.00**
LA19 aug.	5.80	5.93	**5.65**
LA19 cond.	5.00	5.00	**3.87**
DFADD aug.	2.64	2.64	**2.13**
DFADD cond.	**1.44**	1.84	1.60
LibriSeVoc aug.	10.28	10.31	**7.72**
LibriSeVoc cond.	6.33	5.67	**4.33**

aug.: augmented copy (noise, reverberation, codecs); cond.: recorded under a logged condition.

6.2 Source verification and attribution

The same weights verify sources. For two recordings from generators unseen in training (Table 4), *VeriSpoof-Plus* verifies a shared source at 6.17% EER when the generators are new models of known architectures and at 15.00% when the architectures are new, which suggests that source cues are tied to architectures. It names seen generators (96.8%) and recording conditions (81.3%). Degraded copies remain hard (23–30% EER), as noise and codecs mask the detail that separates generators. The pair probe on the frozen DF-Arena-1B encoder reaches only 24.8–29.0% EER on the MLAAD sets; Base is better on every MLAAD, SpoofCeleb and degraded set and worse on STOPA (20.88 vs. 17.9) and ASVspoof 5 (23.27 vs. 22.2), and Plus is better on every set.

General audio LLMs cannot answer these questions. Prompted with the same closed options on 1,000 items per set (Table 5), six general audio LLMs answer the same-source and recording-condition questions near chance (48.4–53.3 and 16.4–22.6% balanced accuracy on average). On authenticity they reach 52.1–64.7%; the best, Qwen3-Omni, reaches 76.0–90.2% on the ASVspoof sets but stays near chance on SpoofCeleb, LibriSeVoc and both CodecFake+ sets. On the same items Base and Plus reach 95.9 and 96.5% on authenticity, 77.9 and 82.8% on same source, and 76.9 and 83.0% on recording condition.

More generators sharpen the fingerprint. Plus improves source verification on 10 of 11 sets (Table 6), by 6–7 points on degraded copies (unseen models 13.80 → 6.17%): seeing more generators sharpens the notion of a source fingerprint. It loses on SpoofCeleb (11.25 → 12.20) and on naming the acoustic model under an unseen vocoder (84.7 → 69.8%), the two tasks that rest on SpoofCeleb's labels, whose share of the training data is far smaller in Plus. Rationale tuning keeps the source answers of Base within 1.3 EER points and 2.4 accuracy points of question tuning, so they survive the last stage through replay.

Table 4. Source tasks. Source verification: EER (%) for same vs. different generator on spoof–spoof pairs (top: clean; middle: augmented / logged-condition copies). Bottom: attribution and recording-condition accuracy (%). Frozen probe: logistic-regression pair probe on the frozen anti-spoofing encoder DF-Arena-1B (learned layer mix), trained on the Base training lists and tested on the same pairs (Appendix C); –: not applicable. Zero-shot audio LLMs: Appendix 5.

Test set	Frozen probe	Base	Plus
MLAAD seen	29.0	17.23	7.68
MLAAD unseen model	26.5	13.80	6.17
MLAAD unseen arch.	24.8	20.25	15.00
SpoofCeleb	21.9	11.25	12.20
STOPA	17.9	20.88	13.55
ASVspoof 2019 LA	13.6	8.22	7.62
ASVspoof 5	22.2	23.27	20.65
MLAAD aug.	31.1	30.33	23.43
MLAAD cond.	37.9	36.70	29.95
STOPA aug.	33.2	29.00	22.97
STOPA cond.	36.5	31.70	25.55
Generator ID acc. (MLAAD seen)	–	95.4	96.8
Acoustic model acc. (unseen vocoder)	–	84.7	69.8
Recording condition acc.	–	75.9	81.3

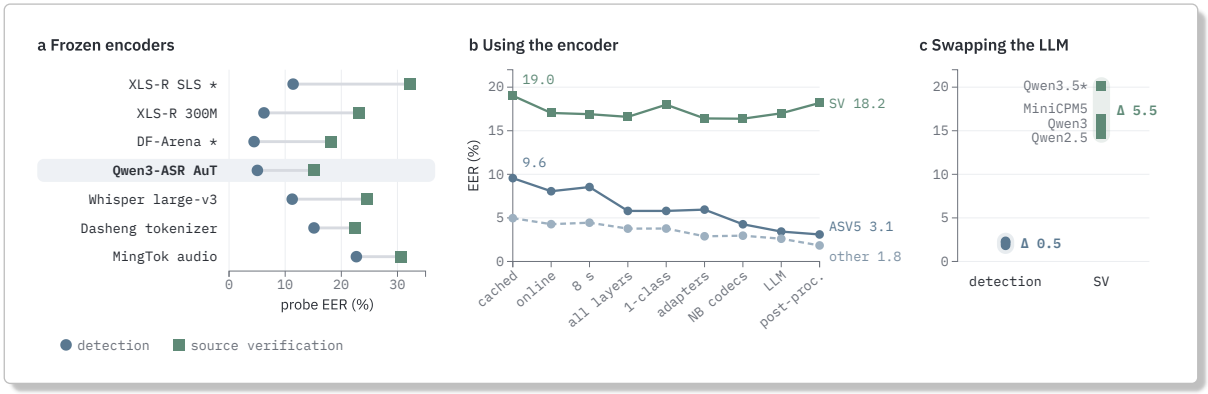


Figure 6. Components. (a) Probes on frozen features of seven encoders (best single layer; same items for every encoder; * trained for anti-spoofing). (b) One factor changed at a time in how the encoder is used, LLM frozen: detection EER on ASVspoof 5 falls threefold, source verification (SV) stays flat. (c) Only the LLM changed: detection moves by 0.5 points, SV by 5.5; * hybrid linear attention. (a) uses an earlier protocol; Table 20 repeats it for two encoders on the current test sets.

Composition depends on the LLM. Naming the acoustic model of SpoofCeleb systems A17 and A18 requires composing an acoustic model seen in training with a vocoder never seen with it (Table 7). Linear probes recover the acoustic model from encoder states for every LLM run (0.84–0.90) and less well from pooled audio tokens (0.64–0.75), yet the answers range from chance (Qwen2.5-1.5B, 0.52) to 0.86 (MiniCPM5-2B, above its token probe). The information reaches the LLM in all runs; whether it is composed into an answer depends on the LLM.

6.3 What determines what the model perceives

An ALM perceives through its encoder, tokenizer and LLM; we ask which of these limits detection and which limits source comparison.

Table 5. Zero-shot general audio LLMs on the *VeriSpoof-Bench*: balanced accuracy (%) on 1,000 items per set drawn from the test items of Table 3 and Table 6, audio as VeriSpoof hears it. Every question lists its options; greedy decoding, the first option named counts (parse rate ≥99.7%). Base and Plus: the most likely answer sentence on the same items. Pairs are presented as one clip, recording A, 1 s of silence and recording B. Qwen3-Omni-30B-A3B [28], MiniCPM-o 4.5 [61], Audio Flamingo 3 [AF3; 29], Kimi-Audio-7B [30], Step-Audio-2-mini [62], MiMo-Audio-7B [63].

Test set	Base	Plus	Qwen3-Omni	MiniCPM-o	AF3	Kimi	Step-2	MiMo
Genuine or spoofed? (chance 50)								
ASVspoof 2021 DF	**99.7**	**99.7**	90.2	76.7	59.9	72.5	55.8	75.4
ASVspoof 2021 LA	94.4	**96.1**	78.0	70.3	59.2	62.5	54.6	72.9
ASVspoof 5	**97.5**	97.1	76.0	75.8	63.9	69.2	54.8	71.6
CodecFake+ CoSG	90.9	**97.6**	50.2	48.6	48.9	57.9	50.2	49.8
CodecFake+	**90.4**	89.7	52.7	59.4	50.1	65.6	50.0	52.2
DFADD	**99.7**	95.9	68.4	65.8	50.1	75.0	50.2	63.0
DFADD aug.	96.3	**97.2**	63.5	56.7	51.0	64.8	51.2	60.7
DFADD cond.	**98.0**	**98.0**	64.0	55.7	48.8	63.6	50.7	59.9
In-the-Wild	**98.7**	97.7	64.0	61.6	49.3	66.6	52.0	58.0
ASVspoof 2019 LA	99.7	**99.8**	77.8	70.1	57.6	63.9	53.6	68.5
LA19 aug.	92.6	**93.4**	70.0	54.9	55.6	64.5	53.9	63.4
LA19 cond.	93.5	**94.9**	69.8	57.0	54.4	65.5	53.6	60.9
LibriSeVoc	**100.0**	**100.0**	52.4	57.9	50.0	55.5	49.8	50.3
LibriSeVoc aug.	89.9	**92.8**	52.8	56.7	49.8	54.7	49.9	49.8
LibriSeVoc cond.	93.4	**94.4**	54.6	58.5	50.5	57.9	49.8	51.1
SpoofCeleb	**99.9**	**99.9**	51.2	58.3	51.0	57.2	53.3	51.5
Mean	95.9	**96.5**	64.7	61.5	53.1	63.6	52.1	59.9
Same source? (chance 50)								
ASVspoof 5	74.0	**80.1**	55.0	52.3	51.4	51.5	50.4	54.2
ASVspoof 2019 LA	91.0	**91.3**	54.5	52.2	50.9	50.7	52.2	50.5
MLAAD seen	85.1	**93.6**	49.4	53.0	48.1	51.1	50.1	52.5
MLAAD unseen arch.	81.1	**86.4**	53.1	58.4	45.2	47.3	50.2	52.3
MLAAD unseen model	86.6	**89.9**	51.4	47.7	46.8	52.1	49.6	49.4
MLAAD aug.	68.9	**76.4**	58.3	57.2	47.9	56.1	50.8	55.5
MLAAD cond.	64.1	**69.6**	48.2	54.3	45.6	53.6	49.9	49.0
SpoofCeleb	87.1	**89.8**	52.5	52.2	48.4	50.6	50.1	54.8
STOPA	80.0	**84.4**	51.6	51.0	50.2	50.7	48.1	52.9
STOPA aug.	71.4	**75.6**	50.2	54.1	48.7	50.1	51.9	52.3
STOPA cond.	67.7	**73.9**	50.6	53.8	48.7	49.6	50.6	50.6
Mean	77.9	**82.8**	52.3	53.3	48.4	51.2	50.4	52.2
Recording condition (5–8 options)								
DFADD aug.	77.0	**86.1**	18.9	17.1	14.2	14.2	18.1	18.4
DFADD cond.	81.7	**86.0**	25.0	22.8	17.3	21.8	16.6	21.9
LA19 aug.	76.5	**80.8**	20.5	15.4	12.8	14.7	19.0	18.0
LA19 cond.	81.5	**85.1**	24.4	21.5	18.1	21.1	17.2	22.6
MLAAD cond.	77.5	**82.2**	22.5	22.2	17.9	21.5	17.1	21.3
STOPA cond.	67.5	**77.7**	24.0	20.9	18.1	18.7	18.6	20.3
Mean	76.9	**83.0**	22.6	20.0	16.4	18.7	17.8	20.4

Table 6. Source tasks of Base after forensic question tuning (QT) and after grounded rationale tuning (final), and Plus (same sets as Table 4): EER (%) of same-source verification on pairs, accuracy (%) in the last three rows. MLAAD seen / unseen model / unseen arch.: pairs of generators seen in training, unseen models of a seen architecture, unseen architectures.

Test set	Base (QT)	Base	Plus
MLAAD seen	17.38	17.23	**7.68**
MLAAD unseen model	14.77	13.80	**6.17**
MLAAD unseen arch.	21.35	20.25	**15.00**
SpoofCeleb	11.50	**11.25**	12.20
STOPA	20.05	20.88	**13.55**
ASVspoof 2019 LA	8.68	8.22	**7.62**
ASVspoof 5	23.52	23.27	**20.65**
MLAAD aug.	29.37	30.33	**23.43**
MLAAD cond.	35.40	36.70	**29.95**
STOPA aug.	28.47	29.00	**22.97**
STOPA cond.	32.20	31.70	**25.55**
Generator ID acc. (MLAAD seen)	96.3	95.4	**96.8**
Acoustic model acc. (unseen vocoder)	82.3	**84.7**	69.8
Recording condition acc.	77.9	75.9	**81.3**

Table 7. Compositional generalization (SpoofCeleb systems A17 and A18: an acoustic model seen in training with a vocoder never seen with it): linear-probe accuracy of the acoustic model on encoder states and pooled audio tokens, and the model's answer (LLM rows after question tuning, as in Table 8(b); Base: final; chance 0.50).

Model	Encoder probe	Token probe	LLM answer
Qwen2.5-1.5B	0.87	0.66	0.52
MiniCPM5-2B	0.84	0.64	0.86
Qwen3-1.7B	0.90	0.75	0.64
Base	–	–	0.85

The speech-recognition encoder keeps the evidence. Probes on frozen features (Figure 6a) show that Qwen3-ASR gives the best source verification of seven encoders and detection close to the best anti-spoofing encoder, while audio tokenizers lose much of both. Repeated under the current protocol against the strongest anti-spoofing encoder, DF-Arena-1B (Table 20), Qwen3-ASR with a learned layer mix verifies sources better (mean EER 24.7 vs. 26.4%, better on 11 of 15 pair sets), and DF-Arena-1B detects better on average (13.3 vs. 14.7%), most clearly on ASVspoof 2021 LA, CoSG and In-the-Wild; the best single layer of Qwen3-ASR is layer 12 of 25 for both tasks. The view that speech-recognition encoders discard the artifacts of synthesis [13] does not hold for this encoder.

Detection depends on how features are used; source identity does not. With the encoder fixed, a ladder of changes in how its features are used (Table 8a, Figure 6b) lowers clean detection EER from 4.97 to 1.84% and ASVspoof 5 from 9.56 to 3.10%. The largest single steps are fusing all 25 encoder layers (ASVspoof 5 8.54 → 5.80), encoder DoRA (degraded detection 8.39 → 6.33) and narrowband codecs in augmentation (ASVspoof 5 5.95 → 4.27). Clean source verification stays between 16.4 and 19.0% throughout. Local, fragile synthesis artifacts gain from varied channels and multi-layer detail, whereas source identity is a global property the features already expose. The last step, post-processing applied to both classes, is the best for detection but raises source-verification EER (17.00 → 18.19 clean); we keep it because it removes the channel shortcuts of §6.4.

Comparison depends on the LLM. With the representation fixed, swapping the LLM (Table 8b, Figure 6c) moves mean detection EER by 0.5 points but clean source verification by 5.5. The hybrid

linear-attention LLM, Qwen3.5-4B [80], is the best detector and the worst comparer, as relating two token sequences needs recall that such attention lacks. Qwen2.5-1.5B [81] has the best clean source verification (14.61%) but states spans poorly (IoU 0.36) and names the acoustic model less often (0.77); MiniCPM5-2B is best on degraded pairs, localization and the acoustic model, and is used for both released models.

Table 8. Components. (a) Representation ladder after perception alignment (LLM frozen; Qwen2.5-1.5B up to R6): each row changes one factor of the row above. R0 trains on Qwen3-ASR features of layers 8 and 11 computed once on 4 s windows; R1 runs the encoder during training, so augmentation is drawn anew for every example; R4 adds the one-class loss and the full ASVspoof 5 training split; R6 adds narrowband codecs to augmentation; R8 applies post-processing to both classes. SV clean / deg.: mean EER (%) of same-source verification over the 7 clean and 4 degraded sets of Table 6; Det. clean: mean detection EER over LA19, ASVspoof 5, SpoofCeleb and CoSG; Det. deg.: over the 6 degraded sets of Table 3; ASV5: ASVspoof 5. (b) Only the LLM changes (question tuning, same recipe; Qwen2.5-1.5B [81], MiniCPM5-2B [35], Qwen3-1.7B [82], Qwen3.5-4B [80]): mean EER (%) over the same SV sets and over the first 8 detection sets of Table 3; localization IoU (HAD, PartialEdit) and acoustic-model (AM) accuracy under unseen vocoders after a rationale-tuning run of the same recipe, which was not run for Qwen3.5-4B (–). Hybrid: linear-attention and full-attention layers. Shaded: the configuration used.

(a) Variant	SV clean	SV deg.	Det. clean	Det. deg.	ASV5
R0 precomputed layers 8, 11; 4 s	19.02	32.17	4.97	9.00	9.56
R1 + encoder in the loop	17.04	33.70	4.28	9.80	8.06
R2 + 8 s window	16.90	33.80	4.45	8.35	8.54
R3 + 25-layer fusion	16.59	31.95	3.77	8.57	5.80
R4 + one-class loss, full ASV5	17.99	33.51	3.78	8.39	5.80
R5 + encoder DoRA	16.41	32.16	2.89	6.33	5.95
R6 + narrowband codecs	**16.38**	31.98	2.96	**5.91**	4.27
R7 LLM → MiniCPM5-2B	17.00	**29.99**	2.61	6.20	3.43
R8 + post-processing, both classes	18.19	33.87	**1.84**	6.62	**3.10**
(b) LLM, question tuning	SV clean	SV deg.	Det.	Loc. IoU	AM acc.
Qwen2.5-1.5B	**14.61**	30.59	2.05	0.36	0.77
MiniCPM5-2B	16.33	**28.98**	2.31	**0.61**	**0.84**
Qwen3-1.7B	15.64	31.36	2.22	0.31	0.76
Qwen3.5-4B (hybrid)	20.15	37.51	**1.85**	–	–

6.4 Channel shortcuts

Without post-processing in training, the model reads channel properties as evidence (Table 9): a 2 kHz high-pass makes 11.6–26.8% of spoofs pass as genuine, and 0.5 s of digital silence or +10 dB gain with clipping makes 12.4 and 15.6% of genuine ASVspoof 5 speech look spoofed. Applying the same post-processing to both classes reduces these to 1.7–8.4%. Labelling the post-processing in the answer gives no consistent further gain (1.9–8.9%), so what removes the shortcut is that both classes share the channel, not that the model is told about it; we keep the unlabelled form. Gain with clipping remains the largest error (8.4%).

6.5 Does knowing the source help detection?

A spoof labelled with its generator says more than "fake", but earlier evidence that source questions help detection did not match authenticity labels or steps [8]. We compare the multi-task model A (Base after question tuning) with two authenticity-only models: B sees as many authenticity questions as A, and B′ trains as many steps and so sees five times more. For each test set and ASVspoof 5 attack we compute $D = \mathrm{EER}_B - \mathrm{EER}_A$ with a paired bootstrap interval and apply a rule fixed before the results: helps (interval above zero), hurts (below), equivalent (inside ± 0.5) or inconclusive.

Table 9. Shortcut probes (perception alignment, 4,000 utterances per set): error rates (%) after a 2 kHz high-pass (spoof judged bona fide; HP), 0.5 s digital-silence padding (Pad) and +10 dB gain with clipping (Gain; bona fide judged spoof), on In-the-Wild (ITW), ASVspoof 5 (ASV5) and ASVspoof 2019 LA (LA19). Labelled: the answer also names the post-processing.

Training	HP ITW	HP ASV5	HP LA19	Pad ASV5	Gain ASV5
No post-processing	11.9	11.6	26.8	12.4	15.6
+ post-processing	3.7	**1.7**	**7.4**	**4.0**	**8.4**
+ post-processing, labelled	**2.2**	1.9	8.9	4.4	**8.4**

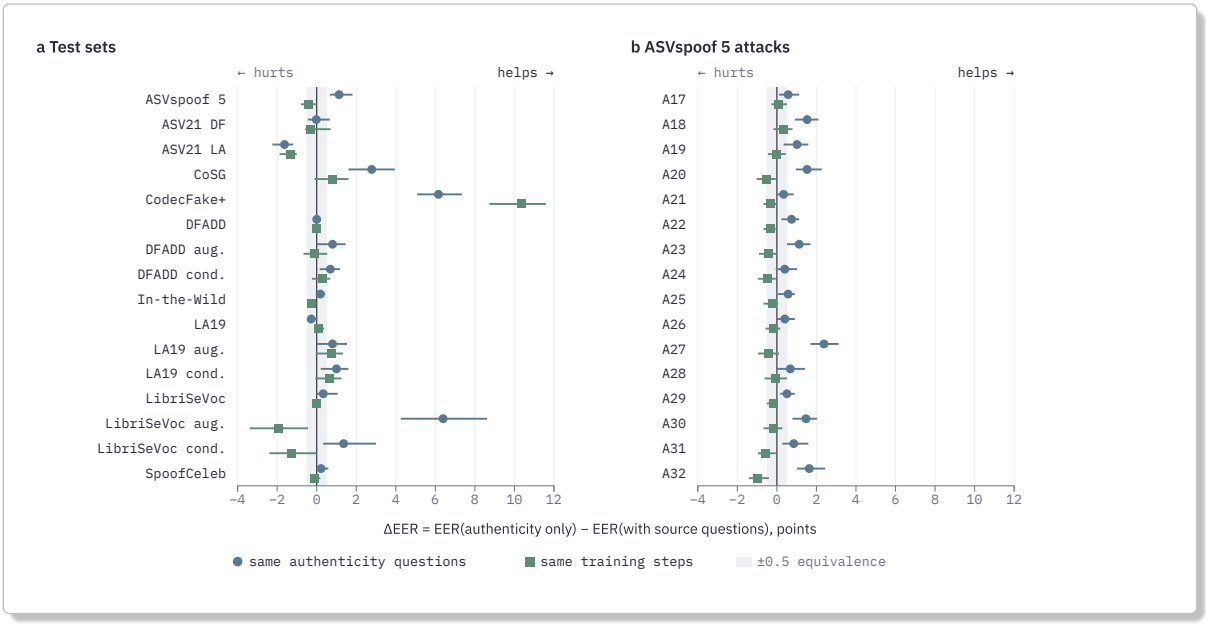


Figure 7. Source supervision per test set (a) and per ASVspoof 5 attack (b; A17–A32 are the attack identifiers of the ASVspoof 5 evaluation set): EER without source questions minus EER with them (95% bootstrap intervals; > 0: source questions help). Circles: same number of authenticity questions (B); squares: same number of training steps (B′); shaded: the ± 0.5 equivalence margin.

Per label it helps; at equal compute it mostly breaks even. Against B (Figure 7, circles; Table 10), source supervision helps on 8 of 16 sets and 12 of 16 ASVspoof 5 attacks and hurts on no attack, most on codec-based generators (CoSG $+2.8$, CodecFake+ $+6.2$) and degraded copies. Against B′ (squares), most differences become equivalent or inconclusive: source labels act mainly by making each example more informative. More binary examples help where the fakes are whole-utterance and the variation is in the channel (In-the-Wild -0.28, LibriSeVoc aug. -1.94 against B′).

For neural-codec fakes it is necessary. With five times more authenticity questions, B′ still has 12.6% EER on CodecFake+ against 2.2% for A ($D = 10.4$ [8.7, 11.6]). Codec resynthesis resembles lossy compression of real speech, so a binary label gives no reason to separate them, whereas asking which codec produced a recording does; this complements the finding that authenticity-only detectors encode generator identity implicitly [2].

An open cost on ASVspoof 2021 LA. ASVspoof 2021 LA is the one set where source supervision hurts (-1.6 against B, -1.3 against B′). The loss sits in three attacks unseen in training (A10, A11 and A18, attack identifiers of ASVspoof 2019 LA that ASVspoof 2021 LA reuses) and grows from question tuning to rationale tuning (3.40 → 5.17%, Table 3); it is not a calibration shift and not tied to one codec. DFADD and LibriSeVoc are at 0% EER for every model and cannot separate the conditions.

Table 10. Source supervision: multi-task model (A) vs. authenticity questions only, with as many authenticity questions as A (B) or as many training steps (B′, about 5× A's authenticity questions). $D = \mathrm{EER}_B - \mathrm{EER}_A$ (> 0: A is better), paired bootstrap (2,000 resamples, stratified by class), 95% interval; decision rule fixed before the results (equivalence margin ± 0.5).

Test set	EER A	EER B	D [95% CI]	Verdict	EER B′	D' [95% CI]	Verdict
ASVspoof 5 (all 30,005)	2.44	3.57	1.13 [0.67, 1.81]	helps	2.04	-0.40 [-0.80, 0.00]	inconclusive
ASVspoof 2021 DF	1.14	1.12	-0.02 [-0.45, 0.66]	inconclusive	0.85	-0.29 [-0.58, 0.70]	inconclusive
ASVspoof 2021 LA	3.40	1.77	-1.63 [-2.24, -1.19]	hurts	2.09	-1.31 [-1.88, -1.01]	hurts
CodecFake+ CoSG	4.51	7.29	2.78 [1.61, 3.95]	helps	5.29	0.78 [-0.11, 1.61]	inconclusive
CodecFake+	2.24	8.40	6.16 [5.08, 7.34]	helps	12.62	10.38 [8.73, 11.59]	helps
DFADD	0.00	0.00	0.00 [0.00, 0.00]	equivalent	0.00	0.00 [0.00, 0.00]	equivalent
DFADD aug.	2.64	3.44	0.80 [0.00, 1.46]	inconclusive	2.53	-0.11 [-0.67, 0.53]	inconclusive
DFADD cond.	1.44	2.13	0.69 [0.15, 1.18]	helps	1.71	0.27 [-0.24, 0.69]	inconclusive
In-the-Wild	1.51	1.70	0.19 [-0.05, 0.43]	equivalent	1.23	-0.28 [-0.48, -0.01]	hurts
ASVspoof 2019 LA	0.47	0.19	-0.28 [-0.46, 0.00]	equivalent	0.57	0.10 [-0.05, 0.37]	equivalent
LA19 aug.	5.80	6.60	0.80 [0.02, 1.53]	helps	6.53	0.73 [-0.05, 1.33]	inconclusive
LA19 cond.	5.00	6.00	1.00 [0.20, 1.60]	helps	5.67	0.67 [-0.07, 1.25]	inconclusive
LibriSeVoc	0.00	0.33	0.33 [0.00, 1.06]	inconclusive	0.00	0.00 [0.00, 0.00]	equivalent
LibriSeVoc aug.	10.28	16.67	6.39 [4.25, 8.61]	helps	8.33	-1.94 [-3.39, -0.44]	hurts
LibriSeVoc cond.	6.33	7.69	1.36 [0.33, 3.00]	helps	5.08	-1.25 [-2.39, 0.00]	inconclusive
SpoofCeleb	0.30	0.52	0.22 [0.00, 0.59]	inconclusive	0.21	-0.09 [-0.28, 0.18]	equivalent

6.6 Every statement can be checked

Table 11. Grounded detection reasoning of VeriSpoof-Base, all measures (500 utterances per set). Acc.: decision after the model's own analysis (%). Agree: agreement with the one-step answer (%). BW (bandwidth class), SNR, Level (speech level), Clip (clipping): share of correct observation statements. IoU of stated spans and F1 of the per-token head on partially spoofed sets. Flip / Para.: share of decisions that follow an opposite localization, in the edit form used in training / a paraphrased form. +proc: copies with post-processing.

Set	Acc.	Agree	BW	SNR	Level	Clip	IoU	Tok. F1	Flip	Para.
LA19	99.6	99.6	0.71	0.97	0.99	0.86	–	0.95	1.00	1.00
ASV5	97.0	98.2	0.77	0.88	0.99	0.73	–	0.98	1.00	1.00
ITW	98.8	98.6	0.64	0.94	0.96	0.99	–	0.99	1.00	1.00
CoSG	97.6	93.8	0.77	0.92	1.00	0.98	–	0.97	1.00	1.00
MLAAD	97.4	87.2	0.72	0.95	0.99	0.99	–	–	1.00	0.96
PartialSpoof	91.2	96.6	0.79	0.98	0.99	1.00	0.25	0.74	1.00	1.00
HAD	99.0	99.6	0.91	0.99	1.00	0.87	0.61	0.91	1.00	1.00
PartialEdit	99.6	78.8	0.68	0.98	0.99	1.00	0.58	0.79	1.00	1.00
LlamaPS	92.2	91.2	0.82	0.94	1.00	0.94	0.31	0.82	1.00	0.99
LA19+proc	99.0	98.6	0.75	0.98	0.99	0.87	–	0.95	1.00	1.00
ITW+proc	98.0	97.6	0.67	0.93	0.95	0.95	–	0.98	1.00	1.00

Most statements are right. Speech level is stated correctly for at least 95% of recordings and SNR for 88–99% (Table 11); clipping is weakest on ASVspoof 5 (0.73), and the bandwidth class is the least reliable statement (0.64–0.91). Probes show that finer bandwidth classes are not recoverable from the audio tokens (38% for 0.5 kHz bins), so coarse classes are the resolution the model can support. A wrong statement fails its check, a signal that free-text explanations lack: in Figure 2 the bandwidth statement fails while the span and the decision hold.

Spans: the frames carry the evidence, writing them is the bottleneck. Stated spans reach IoU 0.58–0.61 on HAD and PartialEdit, where a recording has one or few edited segments, but only 0.25 and 0.31 on PartialSpoof and LlamaPartialSpoof [83], which have many short ones. The per-token head on the same hidden states reaches F1 0.74 and 0.82 there, so the frames are separable and writing many

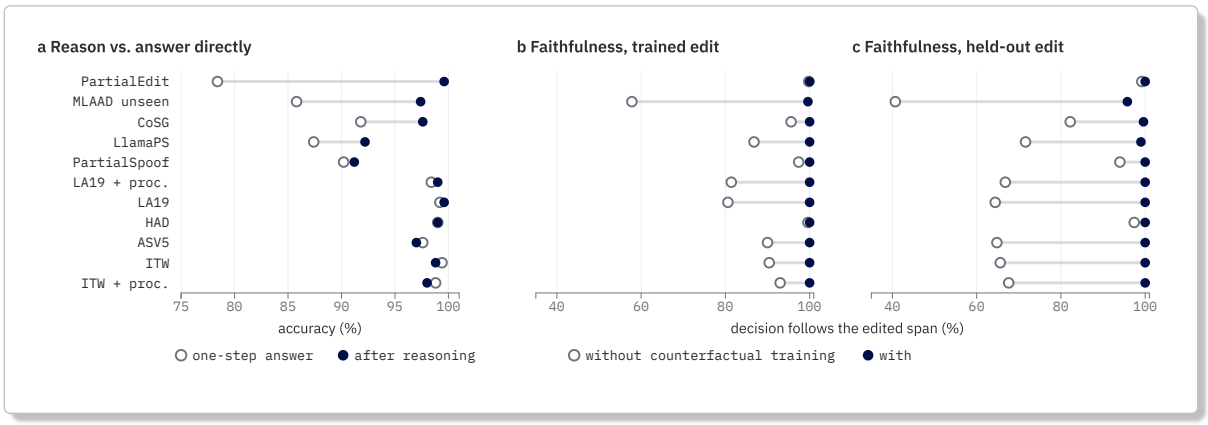


Figure 8. Reasoning and faithfulness. (a) Accuracy of the one-step answer and of the decision after reasoning on the same 500 recordings per set (*VeriSpoof-Base*). (b, c) Share of decisions that follow a localization replaced by its opposite, without and with counterfactual training, for the edit form used in training (b) and a paraphrased form never seen in training (c).

short spans as text is the bottleneck. Span IoU also varies with the LLM (0.25–0.64 on HAD), whereas the per-token head varies little (0.89–0.91; Table 12): the LLM decides how precisely spans are written, not whether the frames carry the evidence.

Table 12. Localization: IoU of the spans stated in the reasoning vs. F1 of the per-token head (synthesized class, 0.16 s grid) on the same utterances of HAD, PartialEdit (PEdit), PartialSpoof (PSpoof) and LlamaPartialSpoof (LlamaPS). LLM rows: the attribute-first rationale-tuning runs of Table 14.

Model	IoU HAD	PEdit	PSpoof	LlamaPS	F1 HAD	PEdit	PSpoof	LlamaPS
Qwen2.5-1.5B	0.35	0.36	0.24	0.30	0.89	0.74	0.73	0.81
Qwen3-1.7B	0.25	0.38	0.19	0.30	0.90	0.76	0.74	0.80
MiniCPM5-2B	**0.64**	**0.59**	**0.27**	**0.31**	**0.91**	0.77	**0.76**	0.81
Base	0.61	0.58	0.25	**0.31**	**0.91**	**0.79**	0.74	**0.82**

6.7 Reasoning versus the one-step answer

Reasoning catches what the one-step answer misses. The decision after reasoning agrees with the one-step answer on 97.6–99.6% of whole-utterance spoofs of known corpora, and where they disagree the reasoning is usually right (Figure 8a, Table 13): 99.6 vs. 78.4% on PartialEdit, 97.4 vs. 85.8% on unseen MLAAD generators. The one-step answer summarizes the whole recording, whereas writing the localization first makes the model inspect it frame by frame. Unlike the chain-of-thought of ThinkOmni, which lowered accuracy [9], our intermediate steps are perception targets with checks.

For Plus the one-step answer is near chance on HAD (48.4%) and 80.0% on PartialSpoof, while its reasoning decision reaches 97.6 and 89.2%: the data added for Plus shift the one-step answer on partial fakes, but writing the localization first still finds the edited span. On whole-utterance sets the one-step answer of Plus catches up (MLAAD 99.4%), and the two agree on 98.4–99.6%. Reasoning is therefore most useful where the training data of the one-step answer do not cover the fake. In practice, the one-step score suits screening at scale, and a rationale suits the recordings an analyst has to decide on.

The order of statements matters. Comparing attributes before the verdict led verification rationales to answer "same source" whenever attributes matched, failing on REVERSAL pairs (accuracy 0.31) although the one-step answer was good (Table 14); in two rationale-tuning runs that differ only in field order, stating the fingerprint verdict first raised this to 0.67. Qwen2.5-1.5B reaches 0.76 even with attributes first, so the failure of the attribute-first order depends on the LLM. The final Base, which states

Table 13. Reasoning decision vs. direct answer on the same 500 utterances per set (%): agreement, accuracy of the reasoning decision and of the direct answer.

Set	Base agree	Base reason.	Base direct	Plus agree	Plus reason.	Plus direct
LA19	99.6	99.6	99.2	99.0	99.0	100.0
ASV5	98.2	97.0	97.6	98.6	98.8	97.4
ITW	98.6	98.8	99.4	99.6	98.0	98.4
CoSG	93.8	97.6	91.8	98.4	98.2	97.8
MLAAD	87.2	97.4	85.8	99.6	99.4	99.4
PartialSpoof	96.6	91.2	90.2	90.0	89.2	80.0
HAD	99.6	99.0	99.0	50.4	97.6	48.4
PartialEdit	78.8	99.6	78.4	94.8	99.6	94.4
LlamaPS	91.2	92.2	87.4	96.2	90.2	88.4
LA19+proc	98.6	99.0	98.4	–	–	–
ITW+proc	97.6	98.0	98.8	–	–	–

the fingerprint verdict first, falls back to 0.42, together with its one-step EER on the same pairs (28.45 vs. 18.35%), which points to its perception of these pairs rather than the format; Plus reaches 0.75. The statement that carries the decision must precede those that could mislead it.

Table 14. Source-verification reasoning: verdict EER (%) after the model's own analysis; REV acc.: accuracy on REVERSAL pairs (different systems sharing every attribute class); Direct: EER of the one-step answer on the same pairs (LA19 cond.). Attribute-first / fingerprint-first: two earlier rationale-tuning runs (MiniCPM5-2B unless stated, without counterfactual examples) that differ only in whether the attribute comparison or the fingerprint verdict comes first; Base and Plus state the fingerprint verdict first.

Format	SpoofCeleb	STOPA	CoSG	LA19 cond.	REV acc.	Direct
Attribute-first	23.40	11.75	24.20	31.70	0.31	19.50
Fingerprint-first	22.80	10.80	26.60	24.95	0.67	**18.35**
Base	20.20	13.65	**21.60**	29.65	0.42	28.45
Plus	**18.40**	14.35	21.80	24.55	0.75	20.20
Attribute-first, Qwen2.5-1.5B	19.40	**10.50**	27.00	**22.75**	**0.76**	20.50

6.8 The decision follows the stated evidence

Without counterfactual training, the model told that no segment was synthesized still answered "spoof" for 42% of unseen MLAAD recordings (Figure 8b); with it, the decision follows the edit on every set. This carries over to edits never seen in training (Figure 8c): a paraphrased opposite localization is followed in 99.5% of cases (73.8% without counterfactual training), an inserted span in 100%, and the decision stays unchanged in 100% when channel or observation statements are contradicted. The decision thus depends on the localization, which is the statement that should carry it, and not on the statements declared non-evidential. The model is never trained on a false statement about the audio, and the statement accuracies of Table 11 are those of the model trained with counterfactual examples.

6.9 Compute: training and running on small GPUs

Only 1% of the parameters are trained. *VeriSpoof* trains 29.7 M of its 2.85 B parameters: the audio tokenizer (17.85 M, all stages), the encoder DoRA adapters (3.24 M, perception alignment), the LLM LoRA adapters (8.60 M, question and rationale tuning) and the per-token head (6 k); a one-class head of 0.53 M is used in training only. The encoder and the LLM stay frozen in bf16 and hold no gradients or optimizer state, so the training state of the adapters (fp32 copy, gradients and Adam moments) takes about 0.5 GB. Memory is set by the frozen weights, about 5.3 GiB in bf16, and by activations.

Training. Table 15 gives the peak memory logged in our runs on one 48 GB L40S. The released Base was trained without gradient checkpointing, which peaks at 21–45 GiB and takes about 14 GPU-hours for the three stages (logged throughput 50, 24 and 18 examples per second). With gradient checkpointing, runs of the same model and stages at the same micro-batch of 32 peak at 11–17 GiB, so the full recipe, global batch included, fits a single 24 GB GPU such as an RTX 3090 or 4090, at a lower speed. Activations grow with the micro-batch, about 0.17–0.34 GiB per example; we estimate that a micro-batch of 8 with gradient checkpointing, accumulated to the same global batch, needs 7–9 GiB per stage, within a 12 GB GPU, and a micro-batch of 1–2 approaches the 5.3 GiB of frozen weights. Offloading the frozen weights to CPU memory, for example with DeepSpeed ZeRO-Offload, would lower the floor further at the cost of speed. These settings change the micro-batch only, not the recipe; we have not trained at them. The code runs in bf16, which needs an Ampere or newer GPU (RTX 30-series, A10, L4 or later); GPUs without bf16 such as the T4 or V100 would need fp16 or fp32, which we have not tested.

Table 15. Peak GPU memory in training (GiB), global batch 32 throughout. Logged on one L40S: the runs of the released *VeriSpoof-Base* (no gradient checkpointing; micro-batch 32 in perception alignment, 8 with accumulation in the other stages), and runs of the same model and stages with gradient checkpointing at a micro-batch of 32. Estimated: micro-batch 8 with checkpointing, from the frozen weights (5.3 GiB), the adapter state (0.5 GiB) and the logged activation memory per example.

Stage	Logged		Estimated
	released Base, no ckpt	micro-batch 32, ckpt	micro-batch 8, ckpt
Perception alignment	45.3	16.8	≈ 9
Forensic question tuning	21.4	11.1	≈ 7
Grounded rationale tuning	24.6	14.4	≈ 8

Inference. All 2.85 B parameters are loaded in bf16, about 5.3 GiB. Scoring one 8 s recording and generating a rationale of a few hundred tokens add well under 1 GiB of activations and cache, so we estimate 6–7 GiB for single recordings: an 8 GB GPU such as an RTX 3060 Ti or 4060 is enough, and about 12 GB covers batched detection (16 recordings per batch by default). Quantizing the LLM to 8 or 4 bits would lower this further, which we have not evaluated. Rationales are the expensive output, as they are generated; the vLLM backend generates them from the same prompt embeddings, with the audio front end in PyTorch (Table 16). Decisions are identical to the reference implementation on 64 of 64 recordings and 32 of 32 pairs; the wording of borderline measurements can differ.

Table 16. Rationale generation time of *VeriSpoof-Base* on one A100 GPU.

Rationale generation	per recording	speed-up
PyTorch, one recording at a time	2.24 s	1×
vLLM, one recording at a time	0.50 s	4.5×
vLLM, batch of 64	0.05 s	46×

7 Conclusion

Speech forensics can be cast as scored question answering with grounded and faithful rationales. *VeriSpoof* reads every answer as a score, writes rationales whose statements can be checked against the signal, and is trained so that its decision follows the evidence it states. With one set of weights it is the best open-weight detector on Speech-DF-Arena and SpoofRadar, verifies the source of recordings from unseen generators, attributes generators and describes recording channels. Controlled studies show where to invest: the representation and how it is trained decide what is detected, the LLM decides how well recordings are compared and how evidence is worded, and supervision that distinguishes how audio was produced adds what more binary labels cannot, at least for neural-codec fakes. Its rationales catch

partial fakes that one score misses and determine the decision even under edits never seen in training. We release both models, the inference code and a demo, and the evaluation protocol for the questions beyond detection.

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A Test sets and licenses

Figure 9 summarizes *VeriSpoof-Bench*; its dataset card lists all 54 subsets with their sources, class balance and the generators or conditions unseen in training.

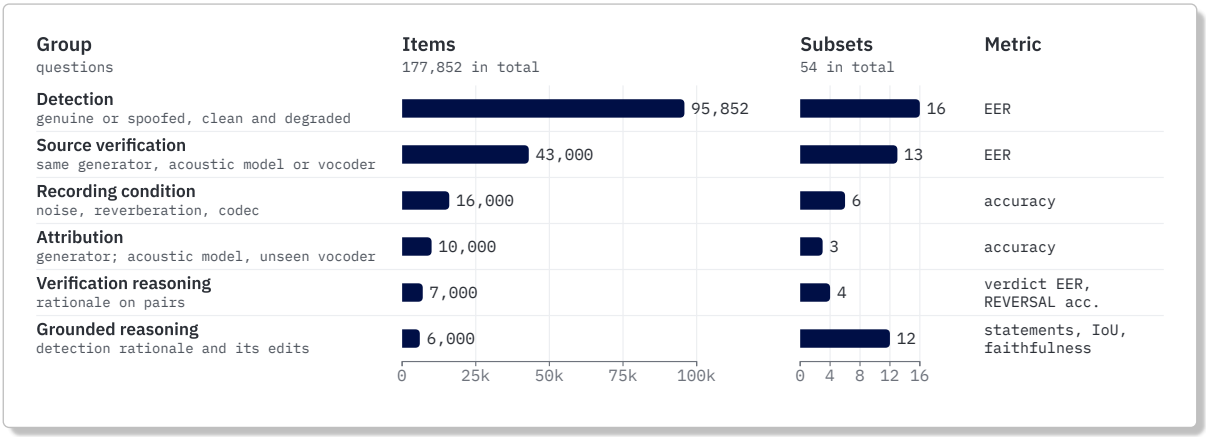


Figure 9. VeriSpoof-Bench by group: items and subsets per group, and the metric of each group. Released with item lists, labels, prompts, scoring scripts and the per-item outputs of both models, without audio.

Table 17 gives the size of every detection test set of the SpoofRadar protocol, whether the training split of the same corpus is used by Base and Plus, and the EER of both models. Table 18 lists the licenses of the corpora in the *VeriSpoof-Bench*. For ASVspoof 2021 DF and In-the-Wild the distributors' statements conflict, and the table gives both. ADD 2022 / 2023 and Codecfake (CC BY-NC-ND 4.0) and Fake-or-Real and SONAR (no license found) are not part of the released lists; they enter only the training data of Plus (ADD 2022, Codecfake, Fake-or-Real) and the comparison on the public protocols. On the 14 common sets alone, Plus has 2.19% pooled EER under the SpoofRadar lists (2.14% in Table 1, which uses the full LibriSeVoc list and 4,528 Fake-or-Real trials).

B Training settings

Table 19 gives the settings of the three stages for both models. Plus runs perception alignment for six balanced epochs over its 211 sources and keeps the step counts of the other two stages.

C Encoder probes

Figure 6a compares seven encoders under an earlier protocol: mean and standard deviation over frames of each layer on a fixed 4.04 s window, detection by logistic regression on up to 8,000 bona fide and 8,000 spoofed utterances per training list, source verification by an MLP on $[|a-b|, a \odot b]$ of 120k training pairs, and earlier test pairs. Table 20 repeats the comparison of Qwen3-ASR with DF-Arena-1B under the current protocol: the Base training lists, the test items of the model evaluation and logistic-regression probes. The frozen-probe column of Table 4 is the DF-Arena-1B pair probe with a learned layer mix.

D Questions and example outputs

Figure 10 shows every question *VeriSpoof* answers, with its template (one of several paraphrases) and an example answer of *VeriSpoof-Base*; the probabilities are those of Figure 3. Figure 11 shows two full outputs with their checks: a PartialEdit fake missed by the one-step answer and caught by the rationale, which also contains a bandwidth statement that fails its check; and a REVERSAL pair decided correctly by the fingerprint verdict although the stated attributes are wrong, an error the checks expose.

Table 17. Detection test sets of the SpoofRadar v1.0 protocol (the 14 common sets, with LibriSeVoc restricted to its test part, plus XMAD-Bench and SpoofCeleb): trials, and whether the training split of the same corpus is used by Base and Plus, and EER (%) of both models; Total row: pooled EER. LA19: ASVspoof 2021 has no training split, and the ASVspoof 2019 LA one is used; ADD22: ADD 2022 train / dev. * AISHELL-3 bona fide speech, 1,001 of the XMAD trials.

Set	Trials	Bona fide	Spoof	Base train	Plus train	EER Base	EER Plus
ASVspoof2019-LA	71,237	7,355	63,882	yes	yes	0.49	**0.22**
ASVspoof2021-LA	148,176	14,816	133,360	LA19	LA19	5.48	**3.44**
ASVspoof2021-DF	533,928	14,869	519,059	LA19	LA19	**0.97**	1.24
ASVspoof5-Eval	680,774	138,688	542,086	yes	yes	**2.59**	3.40
ADD2022-T1	109,199	31,334	77,865	no	ADD22	14.05	**8.65**
ADD2022-T3	126,861	20,776	106,085	no	ADD22	3.87	**1.18**
ADD2023-R1	111,976	80,000	31,976	no	ADD22	5.01	**0.50**
ADD2023-R2	118,477	87,500	30,977	no	ADD22	10.61	**2.41**
Codecfake	304,241	92,596	211,645	no	yes	4.97	**0.12**
LibriSeVoc	18,487	2,641	15,846	yes	yes	**0.00**	**0.00**
DFADD	3,755	755	3,000	yes	yes	**0.00**	**0.00**
SONAR	3,948	2,274	1,674	no	no	5.19	**1.92**
In-the-Wild	31,779	19,963	11,816	no	no	**1.20**	1.90
FoR	4,634	2,264	2,370	no	yes	5.31	**0.09**
XMAD-Bench	102,220	51,110	51,110	no	AISHELL-3*	**0.24**	1.33
SpoofCeleb	91,130	9,113	82,017	yes	yes	0.20	**0.04**
Total	2,460,822	576,054	1,884,768			4.23	4.04

Table 18. Licenses of the internal-suite corpora as stated by each distributor (checked 31 Oct 2026). We release lists, labels and scripts, not audio; each subset's lists carry its corpus' terms.

Corpus	License	Terms for our lists
ASVspoof 2019 LA	ODC-By 1.0	attribution
ASVspoof 2021 LA	ODC-By	attribution
ASVspoof 2021 DF	ODbL (Zenodo record) or ODC-By (asvspoof.org)	attribution; share-alike if ODbL
ASVspoof 5	ODC-By 1.0 (bona fide speech CC BY 4.0)	attribution
In-the-Wild	Apache-2.0 (distributor) or CC BY-SA 4.0 (authors' card)	attribution; share-alike if CC BY-SA
SpoofCeleb	CC BY 4.0, gated	attribution
MLAAD	CC BY-NC 4.0, gated	non-commercial, attribution
STOPA	CC BY 4.0	attribution
DFADD	MIT	none
LibriSeVoc	CC BY-SA 4.0	share-alike
CodecFake+	MIT	none
PartialSpoof, HAD, PartialEdit	CC BY 4.0	attribution
LlamaPartialSpoof	CC BY 4.0	attribution

Table 19. Training settings. PA: perception alignment (tokenizer and encoder DoRA, LLM frozen); QT: forensic question tuning (LoRA and tokenizer, all perception questions); RT: grounded rationale tuning (rationales, counterfactual examples, replay). Task weights are sampling weights. Family: TTS or voice conversion; profile: the attribute classes of one recording's source (generation method, acoustic modelling, waveform generation); authenticity pair: the authenticity of both recordings of a pair; attribute compatibility: whether two recordings from different systems share one attribute class; condition: recording condition. One-class loss: OC-softmax [38] on the pooled audio tokens, pulling bona fide speech towards a learned centre; per-token head: a genuine / synthesized classifier on the LLM's hidden states of the frame tokens.

	VeriSpoof-Base	VeriSpoof-Plus
Encoder	Qwen3-ASR [33], 8 s windows, 25 layers mixed by learned weights, DoRA [34] r=16 (PA only)	
Tokenizer	16 utterance tokens, 52 frame tokens of 0.16 s, time markers every 0.8 s	
LLM	MiniCPM5-2B [35, 36], LoRA [37] r=16, α=32, dropout 0.05 on q/k/v/o	
Losses	cross-entropy; one-class authenticity loss (0.1); per-token head (0.3, RT)	
Steps PA / QT / RT	8k / 20k / 10k	79.2k (6 balanced epochs) / 20k / 10k
Global batch	32	64 (two GPUs, data parallel)
Learning rate PA	tokenizer and DoRA 10^{-4}	same
Learning rate QT	tokenizer 5×10^{-5}, LoRA 10^{-4}	same
Learning rate RT	tokenizer 10^{-5}, LoRA 2×10^{-5}	same
Warm-up	5%	2% (PA), 5% (QT, RT)
Task weights PA	authenticity 3 (Plus 7), family 1, generator 2, same generator 3, recording condition 1	
Task weights QT	authenticity 3, family 1, generator 4, acoustic model 1, vocoder 1, profile 1, same generator 3, same acoustic model 1, same vocoder 1, condition 3, authenticity pair 2	
Task weights RT	detection rationale 5, verification rationale 4, attribute compatibility 1.5, QT replay	
Counterfactual	20% of rationale examples, localization reversed, only the decision trained	

Table 20. Frozen-encoder probes under the current protocol: the Base training lists (120,000 training pairs, 70,744 training utterances), the test items of the model evaluation (8 s windows as the model hears them) and logistic-regression probes on per-layer pooled statistics, with a learned layer mix (mix) or the best single layer (Qwen3-ASR: layer 12 for both tasks; DF-Arena-1B: 8 for pairs, 14 for detection). EER (%). (strata): pairs balanced over SAME, REVERSAL and DIFF.

Test set	Qwen3-ASR mix	best	DF-Arena-1B mix	best
Source verification (pairs)				
ASVspoof 5	25.8	23.7	22.2	**21.5**
ASVspoof 2019 LA	**10.4**	10.7	13.6	15.3
MLAAD seen	**28.4**	30.4	29.0	33.2
MLAAD unseen arch.	**19.5**	22.6	24.8	23.8
MLAAD unseen model	**22.2**	24.6	26.5	31.8
MLAAD aug.	**24.2**	31.3	31.1	32.5
MLAAD cond.	**35.8**	37.1	37.9	40.1
SpoofCeleb	**18.9**	23.5	21.9	22.7
STOPA	22.8	22.3	**17.9**	23.8
STOPA aug.	33.6	**32.4**	33.2	34.8
STOPA cond.	37.8	37.4	**36.5**	39.8
CoSG (strata)	**25.8**	26.4	31.9	32.5
LA19 cond. (strata)	24.7	**21.9**	25.3	26.9
SpoofCeleb (strata)	**23.5**	28.0	25.9	28.3
STOPA (strata)	**17.4**	18.2	18.0	23.0
Mean	24.7	26.0	26.4	28.7
Detection				
ASVspoof 2021 DF	4.2	2.3	**0.8**	**0.8**
ASVspoof 2021 LA	8.7	10.2	**4.1**	7.0
ASVspoof 5	10.4	**9.6**	11.4	11.8
CodecFake+ CoSG	14.9	12.4	6.8	**6.3**
CodecFake+	**1.3**	3.2	3.6	4.5
DFADD	**0.0**	0.1	**0.0**	0.2
DFADD aug.	**34.3**	35.4	36.4	35.9
DFADD cond.	**30.8**	31.1	34.0	32.3
In-the-Wild	6.1	4.8	**2.7**	4.0
ASVspoof 2019 LA	0.6	**0.5**	0.7	1.2
LA19 aug.	**32.4**	36.5	33.3	33.6
LA19 cond.	**27.4**	30.4	29.8	29.4
LibriSeVoc	**0.0**	0.3	**0.0**	**0.0**
LibriSeVoc aug.	35.6	35.4	26.0	**25.0**
LibriSeVoc cond.	27.3	27.0	**22.9**	23.3
SpoofCeleb	0.3	1.0	**0.1**	0.8
Mean	14.7	15.0	13.3	13.5



Figure 10. The questions. Each card gives the question template, an example answer of *VeriSpoof-Base* with the probabilities of its options, and the answer form. One-step answers are read as scores (orange card); rationales are written in fixed fields, abridged here.

Detection rationale, PartialEdit (one word edited, labelled span 1.54–2.05 s); one-step answer: genuine, $P(\text{spoof}) = 0.09$

```
OBSERVATIONS: Reduced bandwidth (about 5-7 kHz); SNR about 30 dB; speech level about -30 dBFS; no clipping.
LOCALIZATION: Synthesized speech at 1.60-2.24 s; the rest is genuine speech. The 3.4 s recording repeats in the window; times refer to its first occurrence.
SOURCE_PROFILE: The generation method cannot be determined.
DECISION: The recording is a spoof: synthetic segments are embedded in genuine speech.
```

Checks: bandwidth fails (measured full band); SNR and level hold; span IoU 0.64; decision correct.

Verification rationale, STOPA REVERSAL pair (different systems, every attribute class shared)

```
CHANNEL_STATUS: The first recording has no audible noise, no noticeable reverberation and Opus compression; the second recording has no audible noise, no noticeable reverberation and no codec compression.
FINGERPRINT: The latent source-fingerprint cues show clear differences.
PROFILE_COMPATIBILITY: Generation method is similar (both copy-synthesis of real speech). Acoustic modelling is similar (both copy-synthesis without an acoustic model). Waveform generation differs (time-domain neural-codec decoders vs GAN vocoders).
DECISION: Channel degradations are present, but they are not evidence about the generation source. Across the source profile, the attributes are mixed. This agrees with the fingerprint cues. Taken together, the recordings come from different generation systems.
```

Checks: decision correct; the attribute statements fail (both systems are diffusion / flow-matching TTS with GAN vocoders), which the check exposes while the fingerprint verdict carries the decision.

Figure 11. Verbatim outputs of VeriSpoof-Base with their checks.